High-Voltage Latching Planar Hall Switch IC

1. Features

- AEC-Q100 Grade 0
- ISO26262 ASIL A
- Latching Planar Hall Switch IC
- Wide voltage range: 2.8V to 40V
- Reverse battery protection: -28V
- Current limited open drain output: 40mA
- Wide operating temperature range:
  - -40℃ to 150℃
- Output short-circuits and overvoltage protection
- High EMC/ESD immunity
- Small package:
  - SOT23-3L (SO)

2. Applications

- Automotive and Industrial
- BLDC motor commutation
- Seat motor adjuster
- Window lifter
- Sunroof/Tailgate opener
- Tachometers

3. Description

SC2498T is manufactured using advanced 60V BCD technology. It is a chopper-stabilized Hall-effect sensor that provides a magnetic sensing solution with excellent sensitivity stability and integrated protection features.

SC2498T employs dynamic offset cancellation, a technique that reduces residual offset voltage typically caused by device overmolding, temperature dependence, and thermal stress. Each device integrates a Hall voltage generator, small-signal amplifier, chopper stabilization circuit, hysteresis comparator, and current-limited output stage.

SC2498T integrates an on-chip voltage regulator, allowing a wide supply voltage range from 2.8V to 40V to meet the requirements of industrial and automotive electronic applications.

SC2498T is packaged in a 3-pin SMD SOT23-3L package (SO) with matte tin-plated leads, and the product is 100% lead-free.



SOT23-3L

Fig.1 Package Outline

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4. Terminal Configuration

SOT23-3L
SO Package
(Top View)

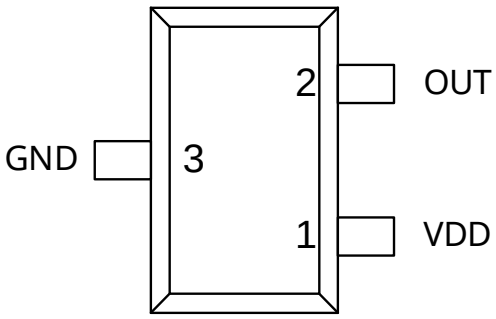


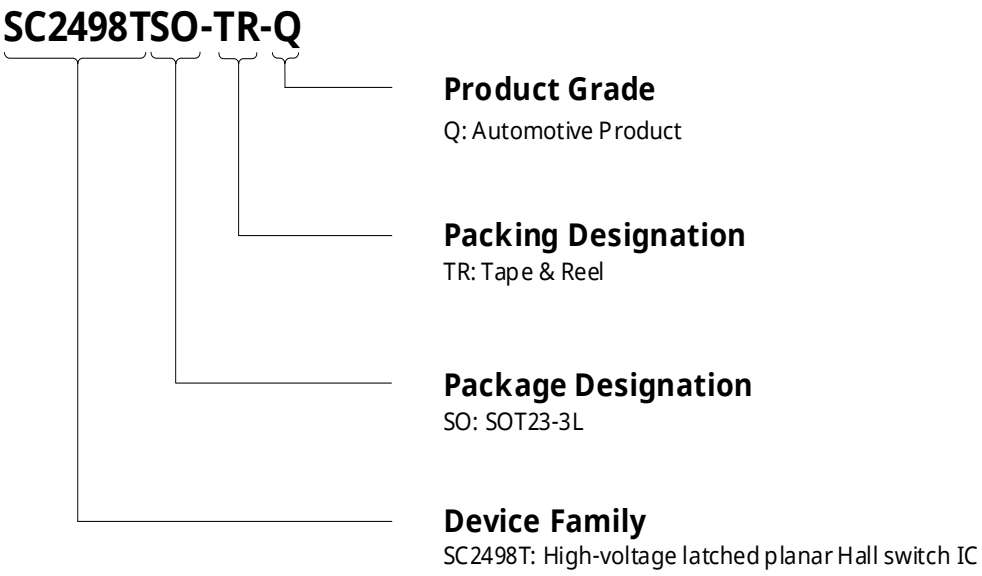
Fig.2 Pin Definition Diagram

Terminal		Type	Description
Name	SO		
VDD	1	Power	2.8V to 40V power supply
GND	3	Ground	Ground terminal
OUT	2	Output	Open-drain output. The open drain requires a pull-up resistor

5. Ordering Information

Ordering Information	Marking	B_{OP}(mT)	B_{RP}(mT)	Ambient, T_A(℃)	Package	Packing	Quantity
SC2498TSO-TR-Q	2498T	3.0	-3.0	-40~150	SOT23-3L	Reel	3000/reel

Ordering Information Format



6. Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted) (1)

Symbol	Parameter	Test Conditions	Min.	Max.	Units
V_{DD}	Power supply voltage	R≥ 200Ω, not exceed 5 min	-28	60	V
V_{OUT}	Output terminal voltage	1.2 kΩ pull up resistor, not exceed 5 min	-0.5	60	V
I_{SINK}	Output terminal current sink		0	40	mA
T_A	Operating ambient temperature		-40	150	℃
T_J	Maximum junction temperature		-40	165	℃
T_{STG}	Storage temperature		-65	175	℃

Note :
(1) Stresses above those listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods may affect device reliability

7. ESD Protection

Symbol	Parameter	Test Conditions	Min.	Max.	Units
V_{ESD_HBM}	HBM	According to: standard AEC-Q100-002 HBM	-8	+8	kV
V_{ESD_CDM}	CDM	According to: standard AEC-Q100-011 CDM	-750	+750	V

8. Thermal Characteristics

Symbol	Parameter	Test Conditions	Rating	Units
$R_{\theta ja}$	SO Package thermal resistance	Single-layer PCB, with copper limited to solder pads	300(1)	℃/W

Note:
(1) Maximum voltage must be adjusted for power dissipation and junction temperature, see Thermal Characteristics

9. Operating Characteristics

9.1. Electrical Characteristics

over operating free-air temperature range V_{DD}=5.0V (unless otherwise noted)

Symbol	Parameter	Test Conditions	Min.	Typ.(1)	Max.	Units
V_{DD}	Operating Voltage (2)	$T_J < T_{J(Max.)}$,	2.8	5.0	40	V
I_{DD}	Operating current	V_{DD}=2.8 to 40V, T_A=25℃,	3.5	4.1	7.0	mA
$UVLO_H$	Under Voltage Lockout High	B>B_{OP}+2.0mT, V_{DD} Rising From 1.5V	2.5	2.7	-	V
$UVLO_L$	Under Voltage Lockout Low	B>B_{OP}+2.0mT, V_{DD} Decreasing From 2.5V	2.3	2.5	-	V
$UVLO_{HYS}$	Under Voltage Hysteresis	$UVLO_H$- $UVLO_L$	0.1	0.2	0.4	V
t_{on}	Power-on time	V_{DD}≥5V	-	25	40	μs
I_{QL}	Off-state leakage current	Output Hi-Z	-	-	3	μA
V_{SAT}	Output saturation voltage drop	B>B_{OP}, V_{DD}=5V, I_O=20mA, T_A=25℃	-	0.14	0.4	V
OCP	Overcurrent protection	Output on $V_{PULL-UP}$<30V	30	40	50	mA
t_d	Output delay time	B=B_{RP} to B_{OP}	-	15	25	μs
t_r	Output rise time (10% to 90%)	V_{PU}=12V, R_{PU}(3)=1kohm C_L=50pF	-	0.2	1	μs
t_f	Output fall time (90% to 10%)	V_{PU}(3)=12V, R_{PU}=1kohm C_L=50pF	-	0.1	1	μs

Note:

(1) Typical values are defined at T_A=25°C, V_{DD}=5V

(2) Maximum voltage must be adjusted for power dissipation and junction temperature, see Thermal Characteristics

(3) R_{PU} and V_{PU} are the external pullup resistor and external pullup voltage

9.2. Magnetic Characteristics

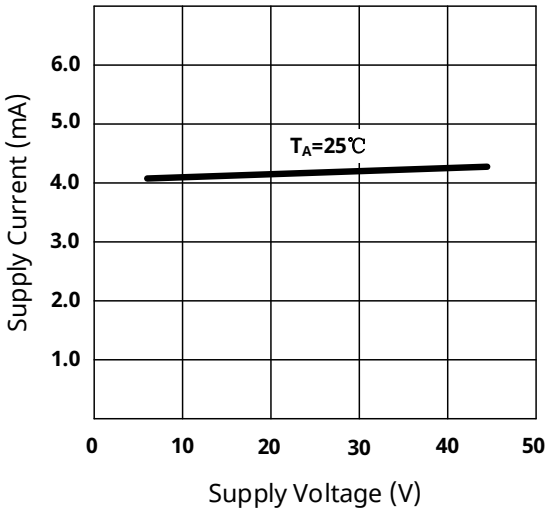
over operating free-air temperature range, V_{DD} = 5.0V (unless otherwise noted)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Units
f_{BW}	BW		20	-	-	kHz
SC2498T 8.0/-8.0mT						
B_{OP}	Operated point	T_A=25℃	1.5	3.0	4.5	mT
B_{RP}	Release point		-4.5	-3.0	-1.5	mT
B_{HYS}	Hysteresis		3.0	6.0	9.0	mT

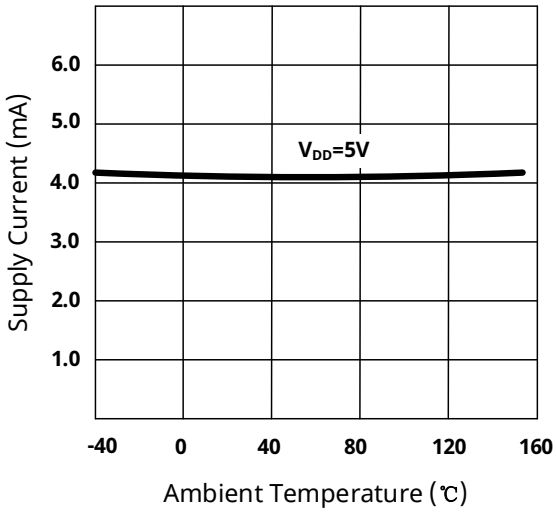
Note:
(1) Magnetic flux density, B is indicated as a negative value for North-polarity magnetic fields, and as a positive value for South-polarity magnetic fields
(2) 1mT=10Gs

10. Characteristic Curves

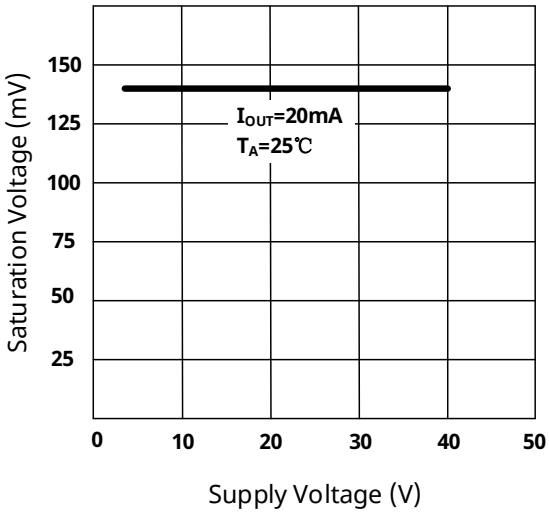
I_{DD} vs V_{DD}



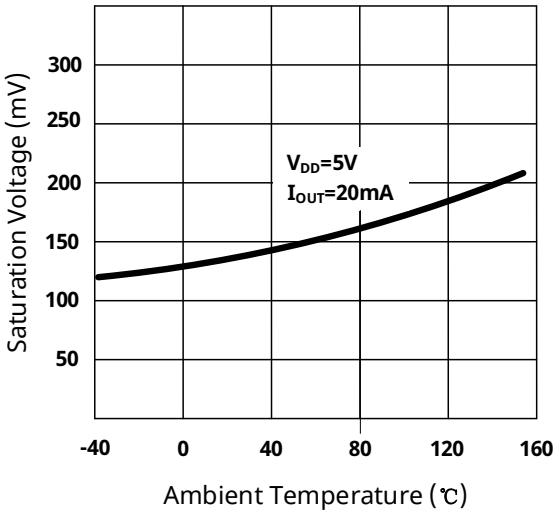
I_{DD} vs T_A



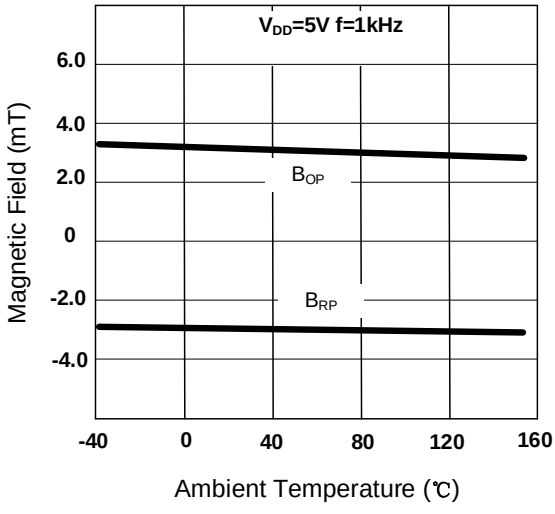
$V_{Q(sat)}$ vs V_{DD}



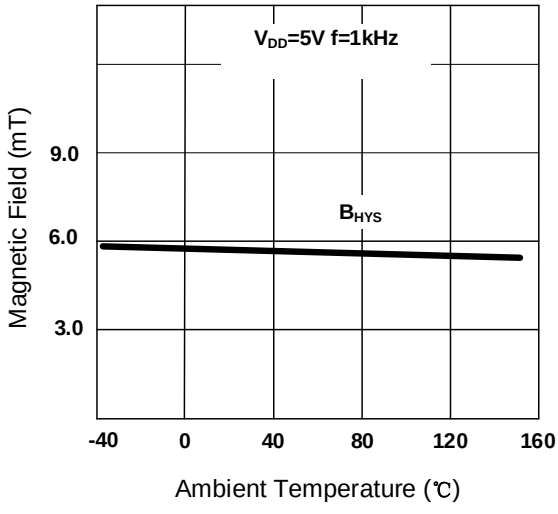
$V_{Q(sat)}$ vs T_A



B_{OP} and B_{RP} vs T_A



B_{HYS} vs T_A



11. Block Diagram

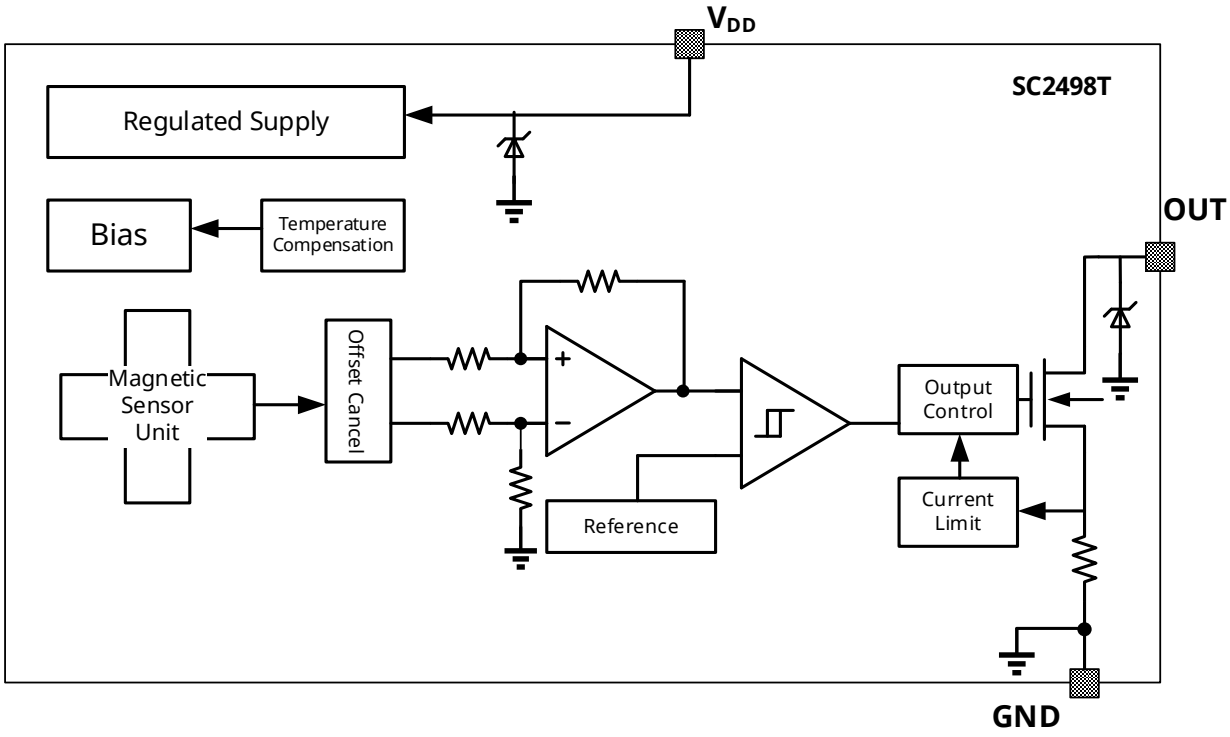


Fig. 3 Functional block diagram

12. Function Description

The SC2498T is a planar magnetic sensing Hall - effect sensor integrated circuit with an open - drain output. The open - drain output is an N - channel metal - oxide - semiconductor (NMOS) transistor that responds to the magnetic field. For the SC2498TT, the direction of the applied magnetic field is parallel to the marked side. These devices are packaged in a small outline (SO) package, which is a 3 - pin surface - mount configuration.

12.1. Magnetic Field Direction Definition

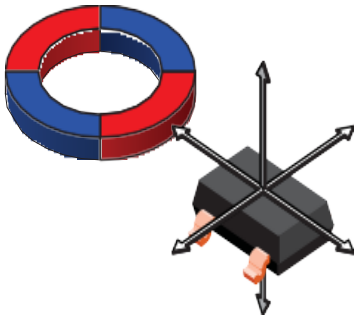


Fig.4 Magnetic Field Direction Definition

12.2. Transfer Function

Powering-on the device in the hysteresis region, less than B_{OP} and higher than B_{RP}, allows an indeterminate output state. The correct state is attained after the first excursion beyond B_{OP} or B_{RP}. If the field strength is greater than B_{OP}, then the output is pulled low. If the field strength is less than B_{RP}, the output is released.

B_{OP}—magnetic threshold for activation of the device output, turning in ON (low) state

B_{RP}—magnetic threshold for release of the device output, turning in OFF (high) state.

$B_{HYS} = B_{OP} - B_{RP}$

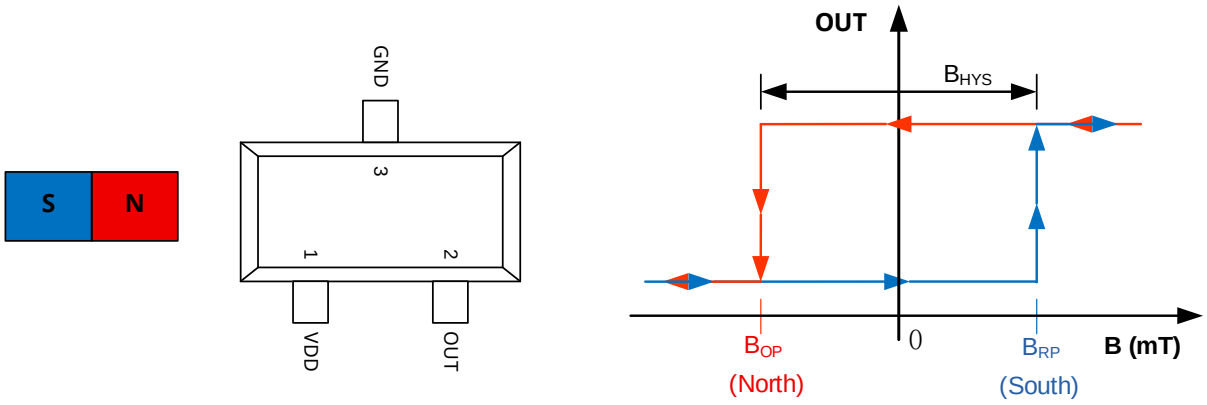


Fig.5 Transfer Characteristic Curve

13. Typical Application

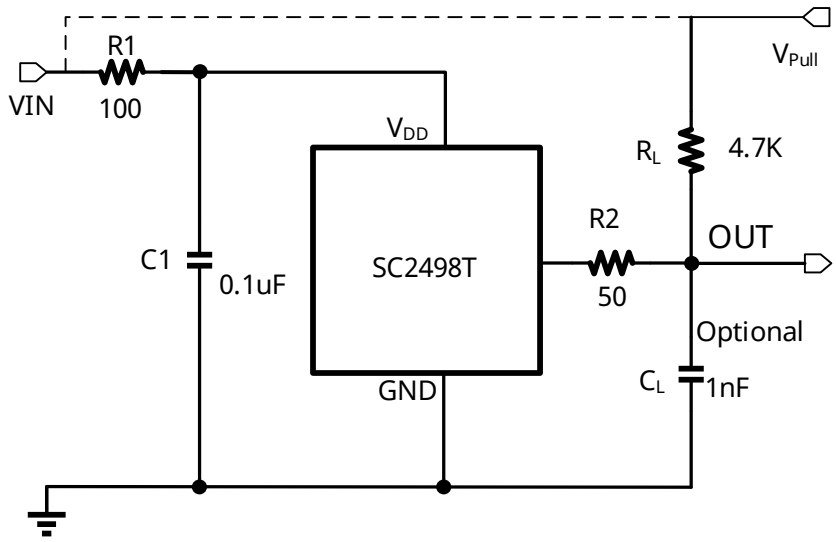


Fig.6 Typical Application Circuit

The SC2498T contains an on-chip voltage regulator which makes the device operate over a wide supply voltage range. In the applications that operate on an unregulated power supply, the external line protection should be added. And the applications using a regulated line, for the EMI/RFI protection purpose, to connect the shunt C_1 capacitors which typically is 0.1μF to the ground near the chip V_{DD} power supply as close as possible and the 100Ω external optional series resistor R_1. The output capacitor C_L used as the output filter is typically 1nF.

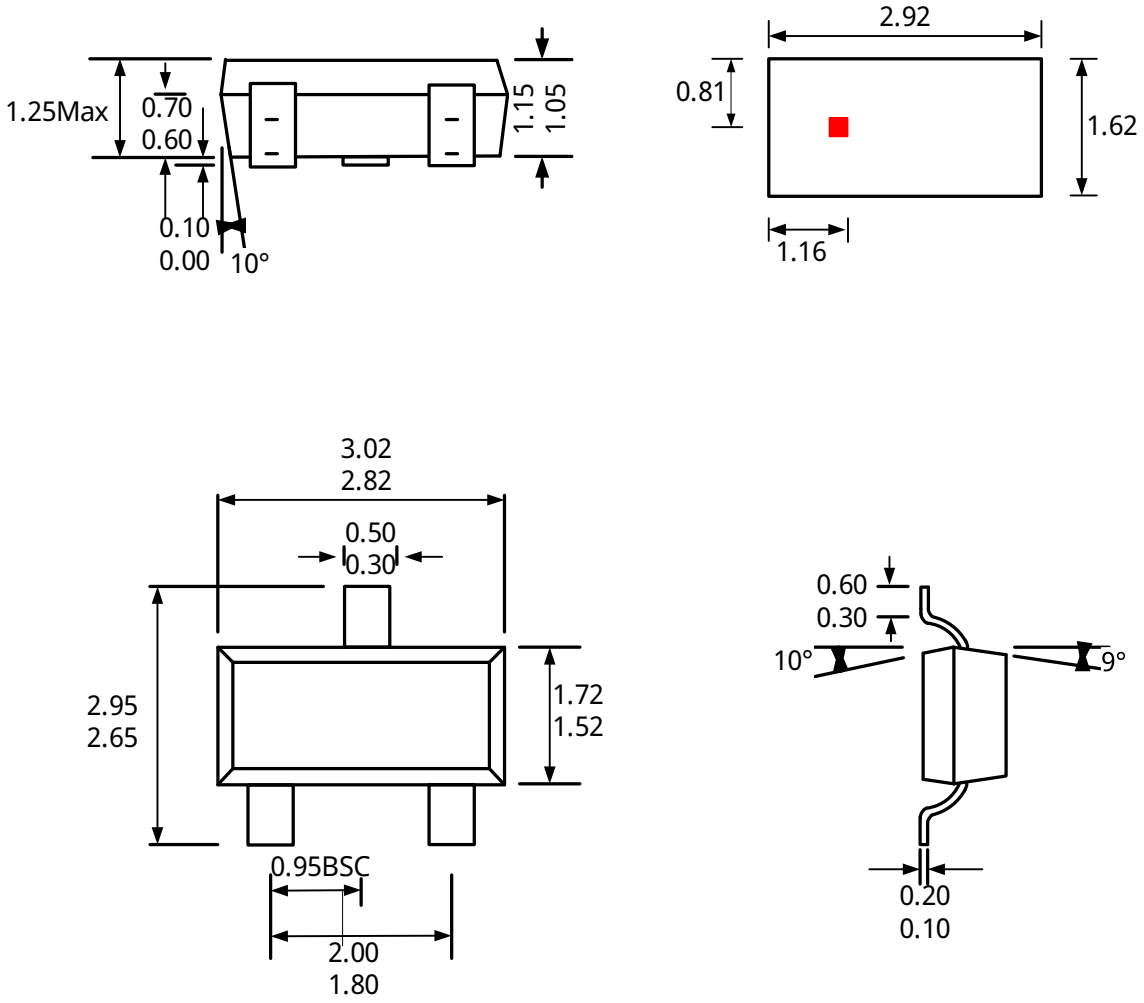
Select a value for C_L based on the system bandwidth specification as follow:

$$C_L < \frac{1}{2\pi \times R_L \times 2 \times f_{BW}(Hz)}$$

V_{PULL} is not restricted to V_{DD} and could be connected to other voltage power supply. The allowable voltage range of this terminal is specified in the Absolute Maximum Ratings.

14. Package Information SO

SOT23-3L Package Outline Dimensions

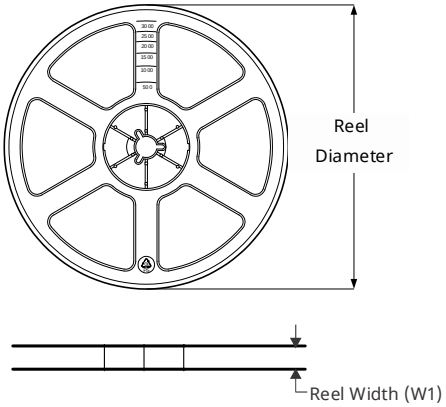


Notes:
(1)All unit in mm.
(2)Dimension does not include mold flash , protrusions or gate burrs.
(3)Allowable dambar protrusion shall be in excess at maximum material condition.

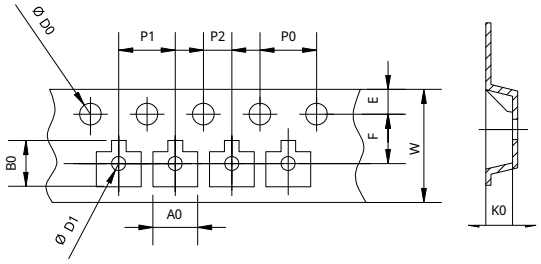
If no tolerance is specified , the dimension shall be theoretical reference value and shall not represent the exact dimension for actual measurement.

15. Tape&Reel Information

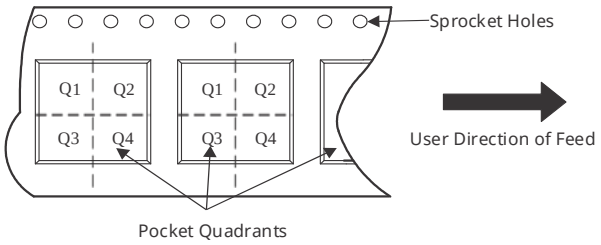
TAPE REEL DIMENSIONS



TAPE DIMENSIONS



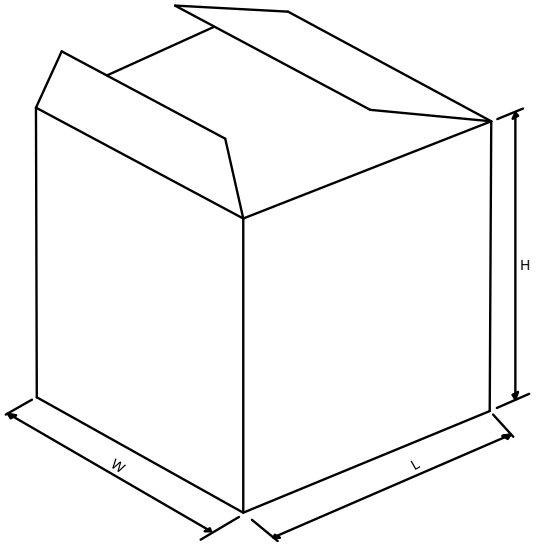
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Package Type	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	P0 (mm)	P1 (mm)	P2 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	W (mm)	Pin1 Quadrant
SOT23-3L	3000	180	8.4	4.00	4.00	2.00	3.18	3.28	1.32	8.00	Q3

16. Tape&Reel Box Dimensions



*All dimensions are nominal

Package Type	SPQ	Length (mm)	Width (mm)	Height (mm)
SOT23-3L	3000ea*10tape	210	210	210

17. Revision History

Revision	Date	Description
Rev.E0.1	2023-06-10	Preliminary datasheet
Rev.E0.2	2023-07-17	Unified format
Rev.E0.3	2023-08-21	Revised some description
Rev.V1.0	2023-08-24	Upgrade version
Rev.V1.1	2024-06-11	Update sensor position
Rev.V1.2	2025-07-05	Release with updated format
Rev.V1.3	2026-05-13	Add packaging information and statements

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